Monotone Classification with Relative Approximations

Lecture Notes for SC2320 Data Analytics and Mining

April 3, 2026

Instructions to readers

The flow of this set of notes is in tandem to the flow of the original paper. It will also go in the same flow as the presentation slides. Concepts/Material/Information that is auxiliary to the core ideas of this paper will be outlined at the end of each section.

Acronym Reference

Acronym	**Full Form**
ER	Entity Resoluton
EM	Entity Matching
ACR3	Acronym Three

1 Background and Motivation

Remark 1.1. Professor Tao's paper itself sometimes uses Entity Matching in the title and introduction to refer more broadly to the problem. For disambiguity in this set of notes, **Entity Resolution** will refer to the entire process while **Entity Matching** will refer to the process that the paper primarily focuses on.

1.1 The Matching Problem

Consider Entities as products in this scenario. Consider two sets of product listings, P_1 and P_2, sourced from two different e-commerce platforms — say Shopee and Lazada. Each listing carries attributes such as `p-name`, `p-description`, `p-year`, and `p-price`. The goal of Entity Matching is to decide, for every pair $(p_1, p_2) \in P_1 \times P_2$, whether p_1 and p_2 refer to the same real-world product.

What makes this non-trivial is that even a genuinely matching pair may disagree on attribute values. For instance, one platform may list a product as *"Sony WH-1000XM5"* while another lists it as *"Sony Noise Cancelling Headphones XM5 Series"*. They are referring to the same product but with different representations. Thus, comparing attributes alone is insufficient. To attain maximum accuracy in entity matching, it falls upon a human expert to manually inspect each pair and make a verdict. However, With thousands of listings on each platform and counting, the number of pairs in $P_1 \times P_2$ grows quadratically - making manual inspection labor intensive and expensive. This motivates the need for a more systematic and efficient approach to Entity Matching.

1.2 The ER Pipeline

To systematically address the Entity Matching problem, the task is transformed into a multidimensional classification problem through the Entity Resolution pipeline. Following the framing of Tao [?], this pipeline consists of three key steps: **Blocking**, **Comparison**, and **Matching**. The first two steps serve as preprocessing stages that progressively reduce and structure the problem, while the third step — **Matching** — is the primary focus of this paper.

1.3 Matching in Depth

1.3.1 Why Human Labelling is Expensive

1.3.2 The Binary Classification Objective

1.3.3 The Monotonicity Rule

1.3.4 Active Classification / Learning

2 RPE Algorithm

2.1 The Algorithm

2.2 Why It Works

2.3 Guarantees and Limitations

3 Coresets (RCC)

3.1 Why RPE Isn't Enough

3.2 The Coreset Technique

3.3 Achieving the $1+\varepsilon$ Guarantee

4 Proofs and Lower Bounds

4.1 Lower Bound Results

4.2 Near-Optimality of the Algorithms

4.3 Mathematical Machinery

Summary

References

[1] Y. Tao, "Monotone Classification," *PODS'18*.

[2] Y. Tao, "Monotone Classification (Extended)," *PODS'21*.